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Financial sentiment analysis meets algorithmic trading: a performance-based approach

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ABSTRACT

This study investigates the use of financial sentiment analysis (FSA) to build trading strategies. We evaluate tweets from 25 large-cap stocks with three linguistic models (BERTweet, RoBERTa, FinBERT), aggregate daily sentiment, and transform it into algorithmic trading signals. The strategies are compared with different buy-and-hold strategies. The results show that sentiment-based portfolios outperform the benchmarks. Despite near-random directional accuracy, the signals capture large movements that improve risk-adjusted performance. Our results highlight the gap between classification metrics and economic value, demonstrating that sentiment can generate alpha under realistic conditions and motivating application-driven evaluation of natural language processing in finance.

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1. Introduction

Sentiment analysis is one of the most widely used and popular methodologies in text analysis. It can effectively extract attitudes, emotions, and opinions about the primary or secondary subject of textual content. For this reason, it is applicable in various contexts where texts or speech transcriptions are available. It has been widely employed in analyzing textual characteristics across diverse domains, including corporate governance (Montone et al., 2024), events (Zhou et al., 2013), specific topics (Thelwall & Buckley, 2013), social issues (Karamibekr & Ghorbani, 2012), entities (Batra & Rao, 2010), politicians (Haselmayer & Jenny, 2017), public institutions (Verma, 2022), and products (Fang & Zhan, 2015). Since the 2000s, the proliferation of websites, the advent of social media, the decreasing cost of access to textual data, and, more broadly, the emergence of big data have significantly driven the growth of this type of analysis. The world of financial markets, intermediaries and banking services has also followed this trend, leading to Financial Sentiment Analysis (FSA) (Du et al., 2024).

In financial markets, sentiment analysis has emerged as a key tool in understanding investor behavior and market dynamics. The textual data used for FSA range from various sources, such as newspaper articles, social media, and financial reports (Chan & Chong, 2017). Following Du et al. (2024), financial sentiment analysis (FSA) can be broadly divided into two main streams: technique-driven and application-driven research. The former focuses on defining tasks, adopting annotated datasets, and advancing methods to improve sentiment analysis performance. The latter emphasizes the practical use of sentiment information in financial markets, particularly in hypothesis testing and predictive modeling. While technique-driven studies concentrate on improving the methodological performance of sentiment

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Public repository: Our source code has been deposited at https://github.com/AlbertoBurchi/fsa_x

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extraction, application-driven research is concerned with the effectiveness and usefulness of sentiment measures within real-world financial decision-making contexts. However, despite rapid methodological progress, there remains surprisingly little evidence on whether high-frequency social-media sentiment can translate into genuinely tradable, risk-adjusted returns once realistic market frictions are considered (Beckmann & Reitz, 2023; Behrendt & Schmidt, 2018). To the best of our knowledge, no prior study has conducted a fully cost-adjusted, out-of-sample back-test of social-media-derived sentiment signals at the individual-stock level. Our study aligns with the application-driven stream, as we investigate the role of financial sentiment in market forecasting and its integration into investment decision-making processes.

We address two research questions: first, can sentiment analysis of Twitter (now X) data predict short-term financial market movements? Second, do sentiment-based trading strategies outperform traditional buy-and-hold approaches? To answer these questions, we employ state-of-the-art language models, BERTweet, RoBERTa and FinBERT, to analyze the sentiment of tweets related to specific stocks. The extracted sentiment scores are aggregated into daily sentiment indicators, which are then used to generate trading signals.

A key novelty of our work lies in the real-world performance evaluation of sentiment-based trading strategies, moving beyond simple predictive accuracy. While previous studies have primarily focused on sentiment classification and its correlation with market movements (Mudinas et al., 2019; Zhang et al., 2023), our approach integrates FSA directly into trading decision-making. Our analysis prioritizes the practical effectiveness of sentiment-driven trading strategies, assessing their ability to generate positive financial performance in real-market conditions. This comprehensive evaluation framework ensures that our findings are not only statistically significant but also actually profitable.

The study contributes to the literature in three ways. First, it evaluates the predictive power of sentiment analysis using alternative advanced models, assessing their effectiveness in financial market forecasting. Second, it provides an empirical comparison of sentiment-based trading strategies, emphasizing financial performance and benchmarking them against alternative investment approaches. Third, the proposed model is applied on a set of listed securities. This work is particularly relevant for both academics and financial practitioners seeking to integrate alternative data sources, such as social media sentiment, into algorithmic trading frameworks.

The rest of the paper is structured as follows. Section 2 reviews the literature on sentiment analysis in finance. Section 3 details the data, sentiment-extraction models, and portfolio-construction methodology. Section 4 presents and discusses the empirical results. Section 5 concludes and suggests avenues for future research.

2. Literature review

2.1. Sentiment analysis in financial markets

Sentiment analysis is now employed in an exceptionally wide range of disciplines. Broadly speaking, existing surveys fall into two groups: (i) topical reviews, which document how sentiment analysis is applied to specific domains, and (ii) methodological reviews, which trace advances in algorithms, corpora, and evaluation metrics (Cui et al., 2023).

Finance has proven to be a natural testing ground for sentiment analysis because the field generates a continuous stream of structured (prices, volumes) and unstructured (text, audio) data. The availability of large-scale corpora and affordable computing has encouraged a rapid expansion of financial-text research over the past decade.

Early work drew heavily on lexicon-based approaches, but recent surveys show a decisive shift toward neural-network and transformer architectures. Mishev et al. (2020) perform a rigorous evaluation of sentiment analysis techniques in finance, comparing the performance of traditional lexicon-based methods with state-of-the-art transformer models. The authors highlight the importance of using domain-specific lexicons for financial texts, as generic sentiment analysis tools often confuse terms in the financial context. There is increasing reliance on diverse data sources, including financial news, analyst reports, social media, and earnings call transcripts, to generate actionable insights (Li & Zhang, 2023).

Birjali et al. (2021) describe a range of methodologies, highlighting their respective strengths and limitations. They highlight key challenges such as language ambiguity, handling domain-specific contexts, and the scarcity of high-quality annotated datasets.

Todd et al. (2024) provide a historical narrative from rule-based dictionaries through deep-learning hybrids. The author emphasizes that the complexity of sentiment analysis techniques has increased over time, leading to more robust sentiment measures. They conclude that domain adaptation, multimodality, and transfer learning are the most promising avenues for future work. Du et al. (2024), arrive at a similar conclusion in their comprehensive review, noting that transformer variants such as FinBERT have already become baselines in market-prediction tasks. They also address the integration of sentiment analysis with other data sources, such as numerical indicators and event-driven models, to enhance predictive accuracy.

Machine learning has been transformative in financial sentiment analysis (FSA) due to its ability to handle complex models in a domain-specific language and diverse data sources. Recently, FSA has been applied to describe investor mood, forecast returns, manage risk, and to power algorithmic trading systems (Du et al., 2024; Guo, 2023; Nixon & Gilbert, 2024). The basis of applying sentiment analysis in trading is the belief that investor sentiment significantly affects stock prices. By capturing market sentiment, the investor could make more informed decisions based on the polarity (Fatouros et al., 2023; Rizinski et al., 2024).

Research on the link between investor sentiment and stock prices has been the focus of many scholars (Cui et al., 2023; Hu & Sun, 2024; Smailović et al., 2014; Xing et al., 2018). Their results suggest that abnormal news sentiment can predict future market returns and volatility, reinforcing the idea that sentiment serves as a valuable trading signal. In trading strategies, sentiment is extracted from various textual data sources and then used as input to operational algorithms aimed at forecasting changes in stock prices (Maqsood et al., 2020; Picasso et al., 2019). The results corroborate the forecasting ability in different contexts of financial markets and economic cycles. Early research focused on news articles and earnings reports, but recent studies have turned to social media data due to its real-time nature. Twitter (now X) has become a valuable source of market sentiment, as it captures public opinion and investor reactions to financial events (Karami et al., 2020; Kumar & Jaiswal, 2020; Swathi et al., 2022; Zou & Herremans, 2022). The advent of big data analytics has further transformed the landscape of algorithmic trading. The study by Almutiri et al. (2022) discusses the application of big data in financial analytics, underscoring its relevance in sentiment measurement and predictive modeling. The ability to process vast amounts of data in real-time allows for more agile trading strategies that can adapt to rapidly changing market conditions.

Recent evidence reinforces this line of investigation. Chang et al. (2022) show that sentiment extracted from stock-message boards predicts one- and two-day returns for small-capitalisation equities. Umar et al. (2021) analyze the GameStop short-squeeze as an extreme case in which sentiment-driven pricing temporarily overwhelms fundamentals. van Eyden et al. (2023) link multi-scale positive and negative price bubbles in G7 equity markets to shifts in aggregate investor sentiment. These studies underscore both the promise and the limitations of sentiment signals, motivating our emphasis on realized trading profitability. Despite encouraging results, the field still faces unresolved issues: domain dependency, the noisiness of short social-media posts, and the trade-off between model complexity and interpretability (Cao, 2023; Heydarian et al., 2025; Man et al., 2019). Moreover, as we argued in the introduction, the literature has yet to establish whether high-frequency sentiment streams retain value after trading costs and real market price dynamics are taken into account.

2.2. NLP and language models for sentiment analysis

Methodologies have evolved from lexicon-based approaches to traditional machine learning models to deep neural networks, and finally, to pre-trained language models (PLMs). Early methods relied on manually curated financial lexicons, such as the Loughran-McDonald Word List, the Henry's Financial Dictionary, the Stock Market Sentiment Lexicon, to extract sentiment from financial texts (Loughran & McDonald, 2011; Oliveira et al., 2016). Machine learning techniques, including Support Vector Machines, Decision Trees, Random Forest, Gradient Boosting and XGBoost, introduced more sophisticated feature

engineering, incorporating sentiment lexicons, numerical indicators, and linguistic features. However, deep learning models revolutionized FSA by automatically learning hierarchical representations of financial text (Sohangir et al., 2018). The effectiveness of FSA using deep-learning neural network models (DNN), specifically Recurrent Neural Networks (RNNs) (Mishev et al., 2020; Yoshihara et al., 2016), Convolutional Neural Networks (CNNs) (Ding et al., 2015; H. Li, 2020) has been proven by many studies. Despite their success, these models required extensive labeled data and struggled with domain adaptation.

A significant breakthrough in FSA came with the adoption of pre-trained language models (PLMs), which leverage transfer learning to enhance sentiment classification accuracy while requiring less task-specific training data (Ergun & Sefer, 2025; Min et al., 2024). Models such as BERT (Devlin, 2018) and its financial-domain adaptations, FinBERT (Araci, 2019; Huang et al., 2023; Z. Liu et al., 2021) and FinGPT (Zhang et al., 2023), have demonstrated state-of-the-art performance in sentiment prediction by capturing complex semantic nuances specific to financial texts. Unlike traditional models, PLMs encode contextualized word representations, allowing them to distinguish sentiment shifts based on financial terminology. For instance, FinBERT, trained on financial reports, analyst discussions, and market news, has been particularly effective in extracting sentiment from earnings call transcripts and financial disclosures (Mishev et al., 2020). Recent advancements, including RoBERTa (Liu et al., 2019) and domain-specific adaptations like BloombergGPT (Wu et al., 2023), further refine sentiment analysis in high-stakes financial applications. The development of advanced language models has significantly improved the accuracy of sentiment analysis. Our study adopts pre-trained language models due to their ability to generalize across diverse financial text sources while maintaining high accuracy in sentiment classification, making them a robust choice for financial market forecasting.

2.3. Stock market prediction

FSA can be practically used in forecasting financial market prices (Li et al., 2020). Turning sentiment into actionable forecasts typically involves three stages: (1) selecting and processing the data source, (2) extracting meaningful features from financial texts, and (3) integrating sentiment information into investment strategies. Each of these steps is crucial in determining the effectiveness and predictive power of sentiment-based market forecasting.

The first step involves the selection and preprocessing of data sources, which has become increasingly complex due to the proliferation of available textual data. Over the years, the variety of sources used for extracting financial sentiment has expanded significantly, making data selection and cleaning essential for reliable predictions. Existing research distinguishes between single-source approaches, where sentiment is extracted from a single type of data (e.g. social media, news articles, financial reports, analyst evaluations), and multi-source approaches, which combine multiple sources to enhance sentiment robustness and reduce noise (Ma et al., 2023). The latter approach is gaining reputation as it allows for a more comprehensive sentiment representation, improving prediction accuracy. In non-English settings, constructing domain- and language-specific resources can be critical for robust signals (Hatefi Ghahfarrokhi & Shamsfard, 2020).

The second step involves feature extraction, where methodologies discussed in Section 2.2 are crucial. Advances in natural language processing (NLP) have significantly impacted FSA, with two future directions emerging: first, the adoption of transformer-based models, which offer improved contextual understanding and financial domain adaptation; second, the integration of multimodal classifiers, which incorporate not only textual data but also audio or visual data to enhance predictive accuracy and move beyond the traditional sentiment classification paradigm (Todd et al., 2024). These advances address some of the limitations of earlier sentiment models by providing more context-aware sentiment insights.

The final step concerns the incorporation of sentiment data into investment strategies, which is as critical as sentiment extraction itself. Simply detecting positive or negative sentiment is insufficient unless it is effectively translated into real investment decisions. Approaches vary in complexity, ranging from basic rule-based strategies, where trades are executed based on sentiment polarity (e.g. buying on positive sentiment, selling on negative sentiment), to more sophisticated models that integrate sentiment as one of multiple decision variables in machine learning-driven trading strategies. These advanced

approaches leverage reinforcement learning, deep learning, and sentiment-aware financial modeling, enabling a data-driven investment framework where sentiment acts as a predictive signal within a broader set of market variables. Consistent with this, Byun et al. (2024) demonstrate that deep-learning signals earn larger alphas precisely for stocks most exposed to behavioural biases.

As research progresses, the challenge remains in optimizing the integration of sentiment analysis within predictive models and trading systems, ensuring that extracted sentiment not only reflects market conditions accurately but also translates into gainful trading strategies.

2.4. Performance measurement in FSA-based market forecasting

The efficacy of FSA applications in stock market prediction depends on one side on the accuracy of sentiment extraction, on the other side on how well sentiment signals translate into profitable trading strategies.

The development of sentiment evaluation methodologies relies on metrics that compare annotated sentiment labels with model-generated sentiment scores across different textual datasets (Luo et al., 2018; Malo et al., 2014; Xing et al., 2020). These methodologies evaluate the performance of FSA models by assessing their alignment with expert-labeled sentiment annotations, using metrics such as F1-score, Weighted Cosine Similarity (WCS), and Matthews Correlation Coefficient (MCC) (Chicco & Jurman, 2020).

Our focus is on how FSA can serve as a tool in the investment decision-making process, providing information by integrating traditional financial analysis. This step is essential so that information extraction does not become a useless exercise. The choice of the evaluation framework is essential to evaluate effectiveness and generalizability: financial markets are dynamic and models must demonstrate both statistical accuracy and profitability in investment activities. Hence, performance evaluation can be based on two approaches: first, predictive performance evaluation; second, trading performance analysis.

The first level of evaluation focuses on the predictive accuracy of FSA models in forecasting stock market trends over different time horizons, typically $t + 1$ (next trading day) or $t + n$ (multi-day forecasts). This allows for the adoption of widely used classification-based evaluation metrics, including: Accuracy, Precision, Recall and F1-Score; which assess the ability of the model to correctly classify stock price movements (i.e. answering the question: 'Will the stock price go up or down?'). However, in financial applications, unbalanced data is common a problem, as upward and downward market movements are not equally distributed. This makes the Matthews correlation coefficient (MCC) a more reliable metric (Chicco & Jurman, 2020).

In recent times, the evolution of techniques involves a change in accuracy measures. Moving towards a no longer dichotomous sentiment, it is also necessary to adopt more complex performance assessments. Mudinas et al. (2019) examine the role of sentiment analysis in stock market prediction, distinguishing between sentiment attitudes (positive/negative) and sentiment emotions (e.g. joy, fear). Their Granger-causality tests reveal that while attitudes do not consistently predict price changes, certain emotions can be predictive in specific cases. Moreover, incorporating sentiment emotions into machine learning models improves stock trend forecasting.

For models aiming to predict stock prices, returns, or movement magnitudes, a regression-based evaluation is more appropriate; thus, metrics such as Mean Square Error (MSE) and R-squared (R^2) can be adopted. MSE quantifies the forecast errors, while R^2 measures the proportion of market variance explained by the model (Wang et al., 2013).

While strong statistical performance in predictive modeling is essential, these metrics are not enough for real trading application. Models need to be validated beyond standard accuracy metrics, especially in real trading scenarios where market noise, macroeconomic conditions and investor behavior introduce non-stationarity.

The second aspect to evaluate focuses on assessing the effectiveness of sentiment-based models in real trading applications. This involves two critical parts: first, the accuracy of sentiment extraction and second, the integration of sentiment signals into the investment decision-making process.

To determine the profitability of sentiment-driven trading strategies, models are often benchmarked through trading simulations. These simulations evaluate whether incorporating sentiment analysis leads to statistically and financially significant improvements in trading performance. To ensure robustness,

performance is often compared against a baseline strategy, typically the Buy-and-Hold (B&H) strategy, which serves as a benchmark to prove the model's ability to generate excess returns. For a comprehensive evaluation, both predictive accuracy and trading performance metrics should be considered, ensuring that sentiment-based models are both statistically reliable and realistically adopted in financial markets.

The previous discussion has presented the approach in a highly schematic manner, which similarly applies to existing research in the literature. Most studies tend to focus on a single aspect of the process, while others attempt to introduce innovations across the multiple steps. However, few works provide a comprehensive, integrated approach that effectively balances methodological advancements with practical implementation.

Ma et al. (2023) propose a Multi-source Aggregated Classification model, which enhances stock price movement predictions by integrating market-driven sentiment, historical price data, and the sentiment extracted from news about related stocks. Their study introduces a graph convolutional network (GCN) to capture the influence of interconnected companies, significantly improving prediction accuracy and trading performance compared to traditional approaches. This study demonstrates the potential of an integrated approach, however it is difficult to attribute the improvement in performance to FSA alone.

3. Data and methodology

3.1. Data collection

Most of the existing literature has focused on assessing the statistical robustness of FSA, while relatively fewer studies have evaluated its practical feasibility within an investment context. In other words, research has largely prioritized forecast accuracy metrics, such as the correct prediction of market direction, over the actual financial performance of trading strategies based on FSA.

To conduct our analysis, we selected two publicly available datasets from Kaggle, both specifically designed for research on sentiment analysis and its relationship with stock price movements:

- **stock_tweets:** a dataset comprising approximately 65,000 tweets related to the 25 most-followed companies on Yahoo Finance, spanning the period from September 30, 2021, to September 29, 2022. Weekend tweets were mapped to the following trading day to align them with the market data and not lose their informative nature by not considering them.
- **stock_yfinance_data:** a dataset containing daily stock prices and trading volumes – open; high; low; close; volume (OHLCV) – for the same set of companies and over the same time frame.

These datasets enable a joint analysis of public sentiment detectable on social media and corresponding market behavior, forming the foundation for our sentiment-based trading strategy.

Regarding the rationale for the choice of data sources; the selection of the Kaggle dataset, the 25-stock universe, and the evaluation window is motivated by the joint objectives of transparency, replicability, and comparability with prior literature. Kaggle provides a publicly available, static, and version-controlled corpus that eliminates the reproducibility issues associated with direct Twitter/X API scraping (where retrospective access is restricted and subject to policy changes since 2023) and has been used in comparable sentiment–stock studies. The 25-stock universe corresponds to the most-followed companies on Yahoo Finance during the sampling period; this filter serves the double purpose of (i) ensuring a tweet volume per ticker sufficient to make daily sentiment aggregation informative and (ii) focusing the analysis on retail-attention names, for which social-media sentiment is a priori more informative than for less-covered securities. The evaluation window (30 September 2021–29 September 2022) covers a complete twelve-month period spanning the post-COVID monetary tightening cycle, the outbreak of the Russia–Ukraine conflict, and a broad equity-market draw-down—that is, a high-volatility, high-attention regime in which real-time sentiment signals are particularly relevant. We acknowledge that generalization to alternative regimes and to non-US equities would require replication on additional samples; this is discussed as a limitation in [Section 4.3](#).

3.2. Sentiment analysis and predictive ability

Sentiment analysis was conducted using pysentimiento, a multilingual Python library designed for social media natural language processing (NLP) tasks. We employed three pretrained models, each fine-tuned for sentiment extraction from English-language tweets:

- a. BERTweet – the default model in pysentimiento, specifically trained on X data;
- b. RoBERTa Large – a robust transformer-based model known for its high performance in general sentiment classification tasks;
- c. FinBERT – a variant of BERT fine-tuned for sentiment analysis in the financial domain, particularly effective for interpreting financial tweets and news content.

These models provide a diverse and complementary set of perspectives for capturing sentiment signals relevant to market dynamics. Using the selected models, tweets were first preprocessed to eliminate noise, including duplicates, special characters, and irrelevant content, ensuring cleaner inputs for sentiment extraction. Each tweet was then assigned a sentiment score consisting of three probability values, reflecting its positivity, neutrality, and negativity.

Because multiple tweets were available for most stocks on a given trading day, tweet-level sentiment outputs were aggregated at the daily stock level. More specifically, for each stock i on day t , the positive, neutral, and negative probabilities were averaged across all tweets associated with that stock on that day. This procedure generated a single daily sentiment profile for each stock, thereby transforming high-frequency textual information into a consistent daily indicator that could be used for forecasting and trading purposes.

To construct an operational trading signal, we assigned each stock-day observation a dominant sentiment label based on the highest average probability among the three sentiment dimensions. A positive daily label was assigned when the positive probability exceeded the negative and neutral probabilities, whereas a negative daily label was assigned when the negative probability was dominant. Neutral sentiment was retained only when neither positive nor negative sentiment was sufficiently informative. This choice was intended to preserve the informational content of the data while reducing the influence of non-directional observations.

Once extracted and aggregated, sentiment was compared with the corresponding stock price movement in order to evaluate predictive ability. Specifically, we tested whether a positive sentiment signal was associated with a positive stock return on the following trading day and whether a negative sentiment signal was associated with a negative next-day return. The prediction target was therefore the direction of the next-day stock movement.

The predictive performance of the sentiment models was evaluated using a set of standard binary-classification metrics: Accuracy, Precision, Recall, F1-score, Matthews Correlation Coefficient (MCC), Specificity, and Negative Predictive Value (NPV). These metrics allow us to assess not only the overall rate of correct predictions but also the balance between true positives, false positives, and false negatives, which is particularly important in financial forecasting environments characterized by noisy and often weak directional signals.

Regarding the rationale for the aggregation rule, the choice to aggregate tweet-level probabilities by arithmetic mean and assign the dominant-class label as the daily operational signal is motivated by interpretability, and consistency with prior literature on aggregation of social-media sentiment at daily frequency. The arithmetic mean does not require calibration parameters and gives equal weight to every tweet, avoiding assumptions about heterogeneous informativeness that would be difficult to validate *ex ante*. The dominant-class rule transforms the three-dimensional daily probability vector into a directly actionable signal for portfolio construction. For tickers with few or no tweets on a given day, the neutral class can prevail mechanically without adding informational content; consistently with the pysentimiento configuration, we therefore apply a fallback that reallocates such days to the second-highest probability class between positive and negative, ensuring that the operational signal reflects the direction of textual evidence when this is available. Alternative aggregation rules (e.g. weighting by tweet-level confidence, restricting the signal to days above a minimum tweet-count threshold, or using probability-margin

thresholds) are possible and can be a natural direction for future work; their implementation would require preserving the tweet-level probability vectors, which are computed as an intermediate step of the pysentimiento pipeline.

3.3. Portfolio construction and trading strategy

Portfolio performance was evaluated by comparing the FSA-based investment strategy with a traditional buy-and-hold benchmark over the same time period. In the sentiment-based portfolio, trading decisions were directly driven by the daily aggregated sentiment scores: a buy signal was created when positive sentiment dominated, and a sell signal when negative sentiment prevailed, both derived from the analysis of daily tweets related to each stock. The benchmark portfolio, by contrast, followed a passive buy-and-hold approach in which each stock was purchased on the first trading day of the evaluation period and then held unchanged until the end of the sample.

The proposed framework is algorithmic in the sense that all trading decisions are generated by predetermined and fully replicable rules, without discretionary intervention. For each stock and each trading day, the sentiment extracted from tweets was aggregated into a daily signal and then translated into a portfolio action. When positive sentiment dominated, the model generated a buy signal. When negative sentiment dominated, the model generated a sell signal. Under the long-only specification, a negative signal was interpreted as an exit or no-position condition. Under the long-short specification, the same negative signal was used to open or maintain a short position. In this way, the methodology provides a rule-based implementation of sentiment-driven algorithmic trading.

The analysis was conducted using *vectorbt*, a Python library for quantitative finance, which enables the efficient implementation and backtesting of algorithmic trading strategies based on user-defined entry and exit rules. Performance evaluation was carried out in two phases. First, we considered a long-only setting, in which the strategy could take only positive exposure to the stock. Second, we considered a long-short setting, in which the strategy could also take negative exposure and therefore potentially benefit from declining prices.

Each portfolio was constructed on a single-stock basis, so that the effectiveness of sentiment signals could be assessed separately for each company in the sample. To reduce look-ahead bias, trades were not executed on the same day on which the signal was observed. Instead, sentiment signals computed using information available on day t were executed at the opening price of day $t + 1$. This design choice ensures that portfolio decisions are based only on information that would have been available to an investor at the time of trading.

To ensure robustness, we also computed average portfolio performance while excluding companies with fewer than 200 tweets, as limited tweet volume may lack the informational richness required to generate reliable sentiment signals. Each portfolio was constructed for a single stock, and to avoid look-ahead bias, trades were executed at the opening price of the trading day following the signal.

The performance of the Sentiment-Based Portfolios was compared with the buy-and-hold benchmark using both return-based and risk-adjusted indicators. In particular, we examined cumulative portfolio returns, the Sharpe ratio, and maximum drawdown. This framework allows us to evaluate whether sentiment extraction contributes not only to directional forecasting but also to economically meaningful trading performance under a clearly defined rule-based strategy.

3.4. Algorithmic trading implementation

In this section, we formalize the algorithmic specification of the implemented trading strategies. Experimentation was conducted in Python using the *vectorbt* library; the complete source code is available in the public GitHub repository.

Signal Generation Rule. For each security i and trading day t , let $s_{i,t} \in \{+1, 0, -1\}$ be the daily sentiment label, where $s_{i,t} = +1$ (positive) when the average probability of the positive class across all tweets published on day t exceeds the average probability of the negative class; $s_{i,t} = -1$ (negative) when the opposite is true; and $s_{i,t} = 0$ (neutral) only when both means are exactly zero. Formally, the aggregate daily probabilities are calculated as the arithmetic mean of the $N_{i,t}$ tweets observed for security i on day

$t: \bar{p}_{ci,t} = (1/N_{i,t}) \sum p_{ci,t,j}$, with $c \in \{\text{pos, neu, neg}\}$. The signal observed at the close of day t determines the target position for day $t+1$: $\text{target_position}_{i,t+1} = f(s_{i,t})$, where f maps the sentiment label to a position based on the trading configuration. In the long-only configuration, $f(+1) = +1$ (long) and $f(0) = f(-1) = 0$ (flat). In the long-short configuration, $f(+1) = +1$, $f(0) = 0$, and $f(-1) = -1$ (short).

Position sizing and capital allocation. Each portfolio is constructed at the individual security level with an initial capital of \$10,000. Positions are unleveraged and non-cumulative: when a signal triggers a change, all available capital is allocated to the new position. Cash held outside of positions does not earn a risk-free rate. Short positions, when permitted, are sized symmetrically with long positions.

Execution and look-ahead bias control. All trades are executed at the opening price of the trading day following the signal. Tweets posted on weekends or non-trading days are mapped forward to the pre-opening window of the following trading day, never backward.

Benchmark strategies. To strengthen the comparative analysis, sentiment-based strategies are compared to the buy-and-hold strategy.

Concerning transaction costs, the results reported in Tables 1–4 correspond to the gross-of-costs configuration ($c=0$ basis points), which isolates the informational content of the sentiment signal from execution frictions. To assess how sensitive the strategies are to realistic transaction costs, we replicate the back-test for $c \in \{0, 5, 10, 20, 50\}$ basis points per trade, applied proportionally to notional at every position trade and to both the sentiment-based and the buy-and-hold portfolios. The full sensitivity results are reported in Appendix A (Table A3).

With respect to robustness at the execution and signal levels, we verify that the reported results are not driven by micro-choices in the execution setup by replicating the back-test under alternative signal-lag and execution-price assumptions. Specifically we consider three configurations: a baseline configuration (signal lag = 1, execution at open of $t+1$), a longer-lag configuration (signal lag = 2, execution at

Table 1. Performance of sentiment-based trading strategies compared to benchmark.

Ticker	Tweet	Benchmark (%)	BERTweet (%)		RoBERTaLarge (%)		FinBERT (%)	
			Long	Long-short	Long	Long-short	Long	Long-short
AAPL	5056	1.70	4.41	3.02	12.42	20.15	15.45	24.07
AMD	2227	−34.22	−38.84	−43.21	−39.73	−44.75	−34.38	−36.98
AMZN	4089	−30.28	−19.95	−12.92	−26.55	−27.20	−3.95	18.99
BA	399	−41.97	−29.70	−17.79	−22.38	−0.31	−21.06	−1.84
BX	50	−29.28	−25.35	−29.02	−14.00	−6.07	−9.32	3.89
COST	393	7.05	−9.55	−25.79	9.72	9.07	−8.08	−23.55
CRM	233	−45.15	−37.45	−29.47	−38.49	−31.48	−50.41	−55.67
DIS	635	−43.08	−27.11	−10.52	−32.00	−21.82	−30.37	−18.13
ENPH	216	91.48	16.85	−42.08	74.04	23.28	56.45	1.16
F	31	−16.90	2.88	25.47	−10.39	−1.06	−26.75	−35.61
GOOG	1291	−26.07	−15.33	−5.83	−14.75	−3.79	−20.32	−16.05
INTC	315	−50.19	−47.99	−45.48	−55.16	−59.45	−54.07	−57.53
KO	310	7.37	16.08	24.49	10.18	12.22	20.75	34.30
META	2751	−59.07	−55.79	−54.96	−58.07	−59.23	−49.65	−43.27
MSFT	4089	−16.39	−10.41	−6.00	−15.67	−17.31	−4.76	3.97
NFLX	1727	−60.26	−52.42	−48.38	−53.45	−50.83	−48.71	−39.76
NIO	3021	−52.76	−44.34	−34.37	−56.31	−58.85	−54.44	−56.54
NOC	31	32.20	30.57	35.65	27.44	29.39	11.97	−1.73
PG	4089	−7.55	−7.38	−9.51	−10.05	−14.39	3.35	12.08
PYPL	843	−65.47	−64.30	−66.80	−65.88	−70.50	−57.43	−53.92
TSLA	37422	8.61	9.77	8.13	21.01	30.99	−4.96	−25.83
TSM	11034	−37.28	−40.64	−44.71	−38.35	−40.35	−21.40	−7.51
VZ	123	−28.24	−19.99	−16.50	−17.92	−12.24	−20.53	−17.62
XPEV	225	−64.17	−35.91	1.42	−38.75	−9.84	−10.23	91.91
ZS	193	−36.76	−31.71	−32.37	−47.45	−58.74	−45.26	−55.16
AVERAGE		−23.87	−21.34	−19.10	−20.02	−18.52	−18.72	−14.25

This table reports the cumulative returns (%) for each stock under sentiment-based trading strategies using BERTweet, RoBERTa Large, and FinBERT, compared to a buy-and-hold benchmark. Results are shown for both long-only and long-short portfolio configurations. Column ‘tweet’ reports the total number of tweets observed for the ticker during the evaluation window (30/09/2021–29/09/2022). ‘benchmark (%)’ is the cumulative return from a passive buy-and-hold strategy executed at the open price of the first trading day and liquidated at the close of the last trading day, expressed as $100 \times (PT/P0 - 1)$. The model columns report the cumulative return of the corresponding sentiment-based portfolio under the ‘long’ (long-only) and ‘long-short’ configurations defined in section 3.4. All returns are gross of transaction costs ($c=0$ bps) and gross of taxes; see appendix A (Table A3) for a full sensitivity analysis with $c \in \{0, 5, 10, 20, 50\}$ bps per trade. ‘AVERAGE’ is the equally weighted cross-sectional mean across the 25 tickers.

Table 2. Performance differential of sentiment-based strategies relative to benchmark.

Ticker	BERTweet (%)		RoBERTaLarge (%)		FinBERT (%)	
	Long	Long-short	Long	Long-short	Long	Long-short
AAPL	2.71	1.32	10.72	18.45	13.75	22.37
AMD	-4.62	-8.99	-5.51	-10.53	-0.16	-2.76
AMZN	10.33	17.36	3.73	3.08	26.33	49.27
BA	12.27	24.18	19.59	41.66	20.91	40.13
BX	3.93	0.26	15.28	23.21	19.96	33.17
COST	-16.6	-32.84	2.67	2.02	-15.13	-30.6
CRM	7.7	15.68	6.66	13.67	-5.26	-10.52
DIS	15.97	32.56	11.08	21.26	12.71	24.95
ENPH	-74.63	-133.56	-17.44	-68.2	-35.03	-90.32
F	19.78	42.37	6.51	15.84	-9.85	-18.71
GOOG	10.74	20.24	11.32	22.28	5.75	10.02
INTC	2.2	4.71	-4.97	-9.26	-3.88	-7.34
KO	8.71	17.12	2.81	4.85	13.38	26.93
META	3.28	4.11	1	-0.16	9.42	15.8
MSFT	5.98	10.39	0.72	-0.92	11.63	20.36
NFLX	7.84	11.88	6.81	9.43	11.55	20.5
NIO	8.42	18.39	-3.55	-6.09	-1.68	-3.78
NOC	-1.63	3.45	-4.76	-2.81	-20.23	-33.93
PG	0.17	-1.96	-2.5	-6.84	10.9	19.63
PYPL	1.17	-1.33	-0.41	-5.03	8.04	11.55
TSLA	1.16	-0.48	12.4	22.38	-13.57	-34.44
TSM	-3.36	-7.43	-1.07	-3.07	15.88	29.77
VZ	8.25	11.74	10.32	16	7.71	10.62
XPEV	28.26	65.59	25.42	54.33	53.94	156.08
ZS	5.05	4.39	-10.69	-21.98	-8.5	-18.4
AVERAGE	2.52	4.77	3.85	5.34	5.14	9.61

This table presents the difference in returns (%) between sentiment-based strategies and the benchmark for each stock. Positive values indicate outperformance. Results are shown for long-only and long-short implementations across the three sentiment models. The performance differential is computed as the arithmetic difference, in percentage points, between the cumulative return of the sentiment-based portfolio and the cumulative return of the buy-and-hold benchmark for the same ticker (as reported in Table 2). Both quantities are computed over the same evaluation window (30/09/2021–29/09/2022) and gross of transaction costs ($c = 0$ bps); see Appendix A (Table A3) for a full cost-sensitivity analysis. ‘AVERAGE’ is the equally weighted cross-sectional mean of the differentials across the 25 tickers.

Table 3. Performance metrics for XPEV and ENPH: Best and worst performing stocks under sentiment-based strategies.

Model	Ticker	Benchmark (%)	Sentiment portfolio (%)	Sharpe ratio	Maximum drawdown (%)
<i>BERTweet</i>	XPEV	-64.17	1.42	0.53	55.77
	ENPH	91.48	-42.08	0.32	66.87
<i>RoBERTa Large</i>	XPEV	-64.17	-9.84	0.36	65.41
	ENPH	91.48	23.28	0.79	67.06
<i>FinBERT</i>	XPEV	-64.17	91.91	1.44	42.61
	ENPH	91.48	1.16	0.51	66.05

This table displays the performances for XPEV and ENPH, identified as the best- and worst-performing stocks, respectively, across all models. The values were calculated, with both long and short positions enabled. ‘Benchmark (%)’ and ‘Sentiment portfolio (%)’ are cumulative returns over the evaluation window (30/09/2021–29/09/2022), gross of transaction costs ($c = 0$ bps; see Appendix A, Table A3, for cost sensitivity). The Sharpe Ratio is computed as $\sqrt{252} \times \text{mean}(rt - rf) / \sigma(rt - rf)$, where rt denotes daily portfolio returns and rf the daily 3-month US T-Bill yield. Maximum Drawdown is the largest peak-to-trough decline of the cumulative-return curve over the evaluation window, expressed as a positive percentage. The extended set of risk-adjusted metrics for XPEV and ENPH (Sortino Ratio, Calmar Ratio, Maximum Drawdown, and Market Exposure) across all three models is reported in Appendix A, Table A1.

Table 4. Average performance of sentiment-based portfolios compared to benchmark.

Portfolio/strategy	Average performance		
	BERTweet	RoBERTa large	FinBERT
Whole portfolio long strategy	2.53	3.85	5.15
Partial portfolio long strategy	1.38	3.97	6.97
Whole portfolio long – short strategy	4.77	5.35	9.62
Partial portfolio long – short strategy	2.84	5.16	13.38

This table summarizes the average performance differentials between sentiment-based portfolios and the benchmark for each of the three models—BERTweet, RoBERTa Large, and FinBERT. Results are reported for both the entire portfolio (whole portfolio) and a filtered subset (Partial portfolio) that excludes companies with fewer than 200 tweets. Each cell reports the equally weighted cross-sectional mean of the per-stock cumulative-return differentials (sentiment portfolio minus buy-and-hold benchmark, in percentage points), computed as in Table 3. The ‘whole portfolio’ row averages over all 25 tickers; the ‘Partial portfolio’ row averages over the subset of tickers with at least 200 tweets in the evaluation window ($n = 16$). All returns are gross of transaction costs ($c = 0$ bps); Appendix A (Tables A2 and A3) reports the corresponding risk-adjusted metrics (sortino ratio, calmar ratio, max drawdown, Market exposure) and the full cost-sensitivity analysis.

open of $t + 2$), and a close-execution configuration (signal lag = 1, execution at close of $t + 1$). Full results are reported in Appendix A (Table A4).

4. Results and discussions

4.1. Sentiment model forecasting performance, statistical evaluation metrics

The predictive performance of the models was evaluated in two stages: first, through classification metrics to assess their ability to forecast market movements; and second, by analyzing the financial performance of the resulting FSA-based trading strategies.

Table 5 reports the statistical forecasting performance of the three sentiment models. Accuracy (ACC) measures the overall proportion of correct predictions. Recall measures the proportion of actual positive return days that are correctly identified by the model. Precision measures the proportion of predicted positive return days that are actually positive. The F1-score is the harmonic mean of Precision and Recall and summarizes the balance between the two. Specificity measures the proportion of non-positive return days correctly classified. Negative Predictive Value (NPV) measures the proportion of predicted non-positive return days that are correct. Finally, the Matthews Correlation Coefficient (MCC) provides a more balanced measure of classification quality in settings where class distributions may be uneven.

The results indicate limited predictive power across all models, with accuracy values hovering around 50%, close to random guessing. This highlights the inherent difficulty of forecasting market movements using FSA alone and underscores the limited reliability of sentiment-based predictions when evaluated solely through classification accuracy.

Recall is notably high for BERTweet (0.8394) and RoBERTa Large (0.8567), suggesting that both models are effective at identifying positive market movements. However, this comes at the cost of low specificity (0.1741 and 0.1507, respectively), indicating poor performance in detecting negative or neutral movements. This can also be generated by a distortion of the news towards positive attitudes. This is due to the models' tendency to recognize instances as positive even when they're not (false positive)

Precision remains low for all models, slightly under 0.50, reflecting a high number of false positives. As a result, while F1-scores are moderately acceptable (around 0.63 for BERTweet and RoBERTa), they still indicate a limited balance between precision and recall.

The Matthews Correlation Coefficient (MCC) values are particularly revealing: all models perform close to zero, with FinBERT slightly negative (-0.0003), suggesting minimal or no correlation between predictions and actual outcomes.

According to the results, FinBERT is the worst model in terms of movements prediction, with better performance on negative instances compared to positive ones. On the other hand, BERTweet and RoBERTa Large performed better in identifying market uptrends.

In summary, while the models are sensitive to capturing upward movements (high recall), their low precision and poor specificity significantly reduce their predictive reliability. These results reinforce the importance of shifting the evaluation toward financial performance metrics, rather than relying solely on classification accuracy. Indeed, standard forecasting metrics do not account for the magnitude of price

Table 5. Forecasting performance metric values of stock market movement based on sentiment detected.

Metric	Forecasting performance		
	BERTweet	RoBERTa large	FinBERT
ACC	0.5041	0.5009	0.4981
Recall	0.8394	0.8567	0.7205
Precision	0.5001	0.4982	0.4960
F1-score	0.6268	0.6301	0.5875
Specificity	0.1741	0.1507	0.2793
NPV	0.5242	0.5165	0.5037
MCC	0.0182	0.0105	-0.0003

Comparison between fine-tuned models of BERTweet, RoBERTa large and FinBERT. The classification target is the sign of the next-day open-to-open return: an 'up-day' (positive class, label = 1) is defined as a strictly positive return, all other days as negative class. Metrics are pooled across all stock-day observations. Definitions: $ACC = (TP+TN)/(TP+TN+FP+FN)$; $Recall = TP/(TP+FN)$; $Precision = TP/(TP+FP)$; $F1\text{-score} = 2 \times Precision \times Recall / (Precision + Recall)$; $Specificity = TN/(TN+FP)$; $NPV = TN/(TN+FN)$; $MCC = [TP \times TN - FP \times FN] / \sqrt{[(TP+FP)(TP+FN)(TN+FP)(TN+FN)]}$. TP, TN, FP, FN denote true/false positives/negatives, respectively. MCC ranges in $[-1, +1]$ with 0 indicating chance-level performance.

movements. For example, a 50 basis point (bps) change is treated with the same importance as a 300 bps change in accuracy calculations, despite the fact that the latter has six times the financial impact in a real portfolio. As a result, forecasting errors should be evaluated not only in statistical terms but also in light of their actual investment consequences, where the size of the misprediction can significantly affect portfolio performance.

4.2. Portfolio performance

Table 1 presents the comparative portfolio performance of individual stocks under sentiment-based trading strategies versus a traditional buy-and-hold benchmark. The evaluation considers both long-only and long-short strategies across three sentiment models: BERTweet, RoBERTa Large, and FinBERT. Table 2 highlights the performance differential (%) of each sentiment-based strategy over the benchmark, under both long-only and long-short configurations. Positive values indicate outperformance relative to the benchmark, while negative values indicate underperformance.

The results show heterogeneous performance across stocks and models, reflecting the complexity of linking sentiment signals to trading decisions. In several cases, sentiment-based strategies significantly outperform the benchmark:

- BERTwet – Long-only strategy average performance improved by 2.53% (80% of portfolio outperformed) and a long-short one by 4.77% (72% of portfolio outperformed).
- RoBERTa Large – Long-only strategy average performance improved by 3.85% (64% of stocks outperformed) and a long-short one by 5.35% (56% of stocks outperformed).
- FinBERT – Long-only strategy average performance improved by 5.15% (60% of stocks outperformed) and a long-short one by 9.62% (60% of stocks outperformed).

If we focus on specific model and stock, the results are more diversified. BERTweet shows moderate outperformance in several cases, notably with F (+42.37%), DIS (+32.56%), and XPEV (+65.59%) in the long-short setup. However, it also demonstrates significant downside risk, particularly with ENPH (−133.56%), suggesting poor alignment between social media sentiment and price action. On average, BERTweet delivers +2.52% (long) and +4.77% (long-short) above the benchmark, modest gains but with high variance across assets.

RoBERTa Large exhibits a more balanced performance, outperforming in most long-short scenarios, especially for BA (+41.66%), KO (+4.85%), and XPEV (+54.33%). Its long-only strategies also generate consistent improvements for stocks like GOOG (+11.32%) and TSLA (+12.40%), while it performs badly with stocks like ENPH and ZS. The average outperformance across all stocks is +3.85% (long) and +5.34% (long-short), indicating a more stable contribution to portfolio returns.

FinBERT clearly stands out in the long-short configuration, showing the highest average outperformance (+9.61%). It achieves exceptional gains in XPEV (+156.08%), AMZN (+49.27%), BA (40.13%). This model shows greater consistency in capitalizing on sentiment signals that translate into profitable trades, particularly for stocks with financial-news-driven sentiment. However, FinBERT underperforms for TSLA, CRM, and NOC. Even for this model ENPH proves difficult to predict with the worst performance.

The table confirms that sentiment-based strategies can outperform passive investing, particularly when short-selling is permitted and when using financially tuned models like FinBERT. However, performance is stock-specific and model-sensitive.

In our analysis, we generated both statistical summaries and graphical comparisons of the sentiment-based portfolios versus the benchmark for each of the 25 companies. Across all three models, XPEV consistently emerged as the top-performing stock, with sentiment-based strategies significantly outperforming the benchmark. In contrast, ENPH consistently underperformed, failing to exceed the benchmark in any configuration and ranking as the worst-performing stock in terms of sentiment strategy effectiveness. Table 3 shows the performance data for these two stocks. Figures 1 and 2 complement the evidence reported in Table 3 by showing the P&L and cumulative return paths for XPEV and ENPH, respectively, under the long-short configuration.

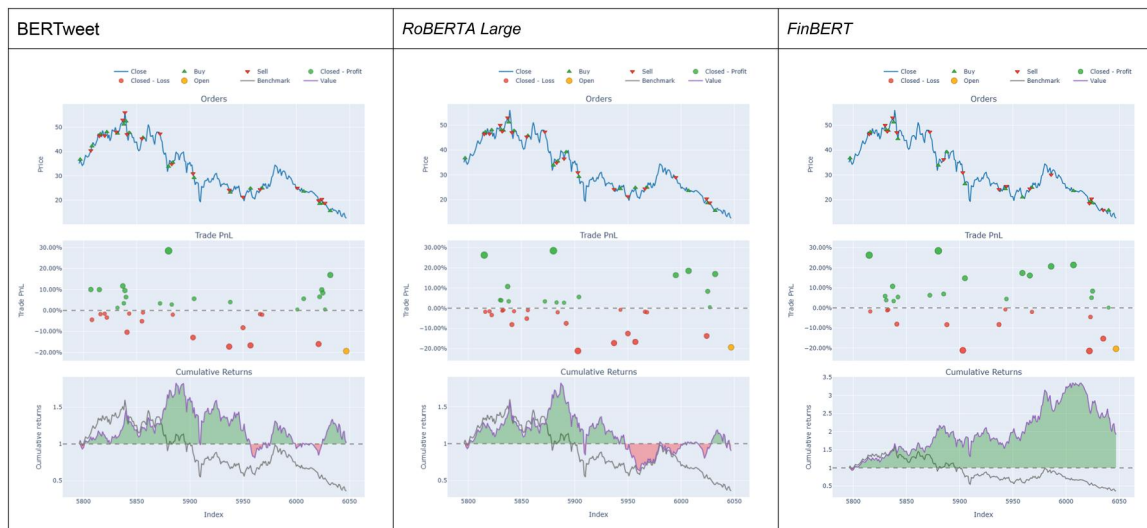


Figure 1. P&L and cumulative returns for XPEV under sentiment-based strategies compared to benchmark. This figure illustrates the profit and loss (P&L) and cumulative returns for XPEV, the best-performing stock in the analysis, using sentiment-based trading strategies with BERTweet, RoBERTa Large, and FinBERT. Results are shown under a long-short trading configuration.

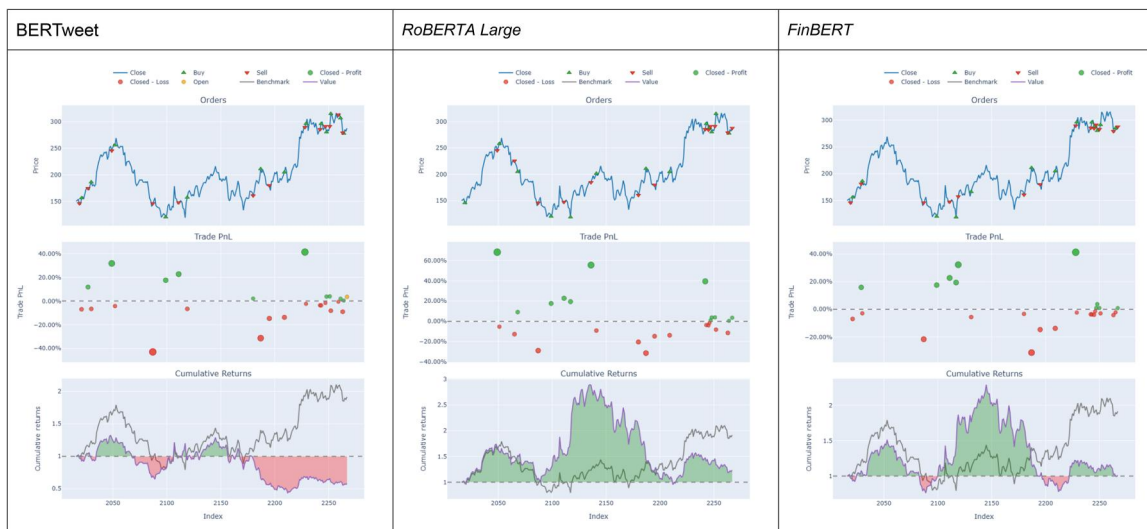


Figure 2. P&L and cumulative returns for ENPH under sentiment-based strategies compared to benchmark. This figure illustrates the profit and loss (P&L) and cumulative returns for ENPH, the worst-performing stock in the analysis, using sentiment-based trading strategies with BERTweet, RoBERTa Large, and FinBERT. Results are shown under a long-short trading configuration.

As shown in Figure 1, XPEV achieved the strongest risk-adjusted performance when FinBERT is adopted as model to assess the sentiment, outperforming the benchmark by an impressive 156%, while also archiving the highest Sharpe ratio (1.44) and the lowest maximum drawdown (42.61%) among all configurations.

In contrast, as shown in Figure 2, ENPH consistently underperformed, with RoBERTa Large demonstrating the most effective handling of its poor sentiment-to-return alignment. While it produced the highest Sharpe ratio for ENPH (0.79) among the models. Additionally, it exhibited a maximum drawdown of 67.06%, reflecting the significant downside risk associated with sentiment-based trading on this stock.

To validate the robustness of our findings, we also evaluated the average portfolio performance excluding companies with fewer than 200 tweets, as limited tweet volume may not provide sufficient information to generate reliable entry and exit signals. Table 4 shows this additional analysis.

Across all models, this filtered subset, referred to as the Partial Portfolio, demonstrated improved results, particularly under the long-short trading strategy. Notably, the FinBERT model, despite exhibiting relatively lower predictive accuracy, delivered the highest average portfolio performance in this

configuration. Its long-short strategy achieved an average return improvement of 13.38% over the benchmark, highlighting the model's effectiveness in translating sentiment signals into profitable trading decisions, especially when supported by a sufficiently rich data setting.

4.3. Discussions

The results of this study provide important insights into the applicability of financial sentiment analysis (FSA) in real-world trading environments. The results show that the predictive performance of models in classifying short-term market movements is limited: accuracy levels are close to random. However, the trading results indicate that sentiment-based strategies can still generate economically meaningful value. This contrast between weak classification performance and stronger portfolio performance is one of the central findings of the paper.

This apparent paradox highlights a key distinction emphasized in the literature: classification accuracy does not necessarily translate into trading profitability, and vice versa (Du et al., 2024; Kazemian et al., 2016; Mudinas et al., 2019). Our analysis confirms that, even when sentiment models struggle to consistently predict the direction of market movements, aggregated sentiment signals can still guide successful trading decisions. This is due to the target variable of the prediction: up-down or long-short.

Binary direction metrics assign the same weight to all forecast outcomes, regardless of the magnitude of the associated price movement. In a trading environment, however, correctly identifying a relatively small number of large price movements may be more valuable than correctly classifying many small and economically irrelevant changes. Positive results likely arise because mispredictions cluster on small moves, whereas correct calls coincide with large price changes. In other words, forecasting error occurs when market volatility is low. Conversely, when the market moves with significant variations, sentiment is a good predictor. These correct predictions have a greater impact on the portfolio than incorrect ones, resulting in a profitable investment strategy. Our findings suggest that the sentiment models may create value not by predicting market direction consistently, but by being informative in periods characterized by larger return realizations.

Among the models tested, FinBERT, although the weakest in terms of predictive classification (with the lowest MCC), consistently outperformed both BERTweet and RoBERTa Large, particularly in long-short configurations. This result suggests that domain-specific language models trained on financial corpora may be better suited to extract economically useful signals than models optimized for more general sentiment classification. This allows for better application in a trading context, even if such signals are not detected by standard accuracy metrics. This observation is consistent with the findings of Mishev et al. (2020; Zhang et al., 2023), who argue that model specialization is more important than overall accuracy for financial applications.

The stock-level analysis further illustrates the heterogeneity of sentiment effectiveness across assets. Stocks such as XPEV, BA, and AMZN demonstrated exceptional alignment between sentiment and trading results. In contrast, ENPH, despite its solid performance in the benchmark, showed weak or even inverse correlations with sentiment signals, regardless of the model used. The difference does not appear to be due to the number of messages on which sentiment was calculated. This result underscores the need for stock-specific calibration, as sentiment-based strategies are not universally effective. Furthermore, a prior analysis of message quality appears appropriate. On the one hand, this introduces an additional step into the methodology that must be managed to avoid subjective approaches and identify appropriate settings. On the other hand, it could exclude messages that, rather than accumulating information, might lead to greater disturbance.

The robustness analysis also provides an important insight. When stocks with fewer than 200 tweets are excluded, portfolio performance improves across all three models, especially in the long-short configuration. This finding supports the view that data density matters. In noisy social-media environments, sparse textual coverage may amplify randomness rather than provide informative signals. A sufficiently rich volume of messages appears to be an important condition for turning sentiment extraction into an effective trading input.

At the same time, the results should be interpreted in light of several limitations. First, the analysis covers a single one-year period, from September 30, 2021, to September 29, 2022, a timeframe

characterized by specific macroeconomic and geopolitical events. The economic crisis caused by COVID-19 is not far off, and its effects are still being felt. These conditions created an environment in which market sentiment has been highly reactive, particularly on social media platforms. While this volatility increases the relevance of using real-time sentiment signals for short-term trading, it also introduces noise that can impact model performance. While the results are encouraging, the potential of sentiment-based trading strategies may vary with changing market conditions and should be appropriately accounted for in the model.

Second, the 25-stock basket consists of the most followed companies on Yahoo Finance, which biases the sample toward large-cap US stocks of interest to retail investors, where social media sentiment is likely more informative than less ‘popular’ stocks. The study should be extended to small-cap and mid-cap or non-US stocks.

Third, although signals are calculated from tweets posted up to and including day t and acted upon at the open of day $t + 1$, tweets posted overnight between the close and the open of day $t + 1$ may already incorporate information released after the close.

Fourth, Twitter (now X) is a noisy and demographically biased source of investor sentiment; the sentiment captured does not necessarily reflect institutional positioning or aggregate market sentiment. Furthermore, two of the three models (BERTweet and RoBERTa Large) were optimized on general-purpose corpuses rather than financial texts, which is consistent with their tendency to overclassify positive sentiment and may explain part of the performance gap compared to FinBERT, which is optimized for the financial sector.

The sensitivity analysis in Appendix A (Table A3) shows that the average outperformance of the sentiment-based strategies relative to the buy-and-hold benchmark is preserved for transaction-cost levels up to approximately 5 basis points per trade, with FinBERT in the long-short configuration maintaining a positive differential of about +5.2 percentage points at $c = 5$ bps. At $c = 10$ bps the average differential approaches zero for all three models (roughly +1.0, −0.1, and −1.3 percentage points, respectively, for FinBERT, RoBERTa Large and BERTweet in the long-short configuration), and at higher cost levels the strategies underperform the benchmark. We interpret this pattern as evidence that the sentiment signal carries information that is economically meaningful in the informational sense but that becomes marginal under execution frictions. This has an important methodological implication: the contribution of this study is to demonstrate that daily social-media sentiment—particularly when extracted with domain-specific transformers such as FinBERT—contains directional information that yields risk-adjusted improvements over passive benchmarks in the absence of frictions, thereby justifying further work on execution optimization. The ranking of the models (FinBERT > RoBERTa Large > BERTweet) and of the trading configurations (long-short > long-only) is preserved across all cost levels tested.

Appendix A (Table A4) reports the average differential of each (model, configuration) pair under three alternative execution and signal specifications: a longer signal lag (execution at $t + 2$ rather than $t + 1$) and execution at the close of $t + 1$ rather than at the open of $t + 1$. Across all scenarios, the sentiment-based portfolios maintain a positive average differential against the benchmark, and FinBERT in the long-short configuration remains the top performer in the baseline and in the close-execution scenarios, while its advantage narrows under the longer lag—consistent with the idea that the sentiment signal decays rapidly and is best exploited with minimal execution latency. The relative ranking of models and configurations is preserved.

Appendix A (Tables A1 and A2) reports the extended set of risk-adjusted metrics (Sharpe, Sortino, Calmar, Maximum Drawdown, Market Exposure) for the two representative stocks (XPEV, ENPH) and for the whole and partial portfolios. In line with the cumulative-return evidence, FinBERT in the long-short configuration delivers the best risk-adjusted performance: on the whole portfolio, average Sharpe is −0.26 (against −0.64 for the buy-and-hold benchmark), average Sortino −0.29, average Calmar −0.09, and market exposure 99.0 per cent. The negative sign of Sharpe ratios reflects the fact that the twelve-month evaluation window is a bear-market phase for the reference universe; the correct interpretation is that the sentiment-based portfolios are *less negative* than the passive benchmark, i.e. they compress downside risk on a per-unit-of-volatility basis. On the partial portfolio (tickers with at least 200 tweets), FinBERT in the long-short configuration further improves to an average differential of +13.4 percentage

points and an average Calmar of +0.02, consistent with the intuition that sentiment signals are more informative when built on denser textual data.

Despite the above caveats, the consistent qualitative pattern found across securities, models, and benchmarks supports the conclusion that sentiment signals convey economically significant information beyond what classification accuracy alone would suggest. The main practical implication is methodological. The study supports a practical interpretation of FSA in finance. Practitioners and researchers evaluating sentiment-based strategies should rely primarily on risk-adjusted financial metrics rather than directional accuracy and should validate strategies based on realistic assumptions about transaction and execution costs.

5. Conclusion

This study evaluated the use of financial sentiment analysis (FSA) for short-term stock market forecasting and its application in algorithmic trading strategies. Although the predictive performance of sentiment models in classifying the direction of next-day market movements was modest, the results clearly show that sentiment-based algorithmic trading can generate superior value compared to a traditional buy-and-hold approach. Performance further improves when short positions are also allowed and domain-specific models (specifically, the financial sector) such as FinBERT are applied. Thus, this study highlights the potential of pre-trained linguistic models applied to microblogging for sentiment extraction.

The empirical evaluation of how sentiment signals can be used in practical trading emphasized financial performance metrics, such as the Sharpe ratio and drawdown, over mere classification accuracy, thus filling a key gap in the current literature. The results highlight the importance of model selection and target variables during backtesting. Furthermore, it is important to consider the specific behavior of the security and the specificities of the market moment.

Future research could build upon this work in several important directions. First, applying large language models, which offer advanced contextual understanding and reasoning capabilities, could further improve sentiment prediction and the interpretation of complex financial text. Second, extending sentiment analysis to cryptocurrency markets (Han, 2025), where investor sentiment often plays a huge role in price dynamics. Finally, combining sentiment signals with alternative data sources, such as web search trends or ESG scores, could lead to more robust and adaptive trading strategies that capture a broader range of phenomena.

Finally, our results demonstrate that sentiment analysis can improve algorithmic trading strategies, even when the accuracy of directional forecasts is modest. Indeed, although models may not reliably predict daily price direction, they effectively capture large price variations, allowing strategies to capitalize on significant movements. These findings highlight the potential of integrating natural language processing (NLP) techniques with financial analysis. Sentiment-based strategies offer a valuable complement to investment approaches, especially in volatile and information-sensitive markets. However, to fully realize their potential, challenges such as data noise and domain dependency must be addressed in future model research.

Author contributions

CRedit: **Alberto Burchi**: Conceptualization, Funding acquisition, Investigation, Methodology, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Leonardo Regni**: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft.

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Data availability statement

The data were obtained from publicly available sources. Specifically, the tweet data used in the analysis come from the Kaggle dataset 'Stock Tweets for Sentiment Analysis and Prediction'. The source code used for data processing, price downloading, sentiment extraction, and back testing is available in the authors' public GitHub repository (https://github.com/AlbertoBurchi/fsa_x). Any derived data files and additional materials supporting the findings of this study are available from the corresponding author upon reasonable request.

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